

paper_01_physics_engine

Paper 1 Brief: A Symplectic Physics Substrate for Autonomous Agent Simulation

Authority state: INTERNAL_RESEARCH

Claim boundary: `partial_repo_verified`; current repo verifies a tick-based 3D simulation layer, while Velocity Verlet/N-body/relativistic claims remain evidence-gated.

Abstract

This paper presents the numerical mechanics layer of *Spatium Computationis v3*, a proposed substrate for modeling autonomous agents as physically situated entities. The paper focuses on Velocity Verlet integration, pairwise gravitational and electromagnetic interactions, relativistic speed bounds, elastic collision handling, and conservation-law diagnostics. Its central claim is not that the system is fully validated in the supplied material, but that a symplectic mechanics layer provides a disciplined foundation for agent simulation when paired with measurable energy, momentum, angular-momentum, and causality checks. The paper defines the substrate, identifies its numerical invariants, and specifies the evidence required for public validation.

Distinct Thesis

A symplectic physics engine can serve as the first layer of a physically grounded autonomous-agent substrate by making agent interaction, influence, collision, and propagation constraints measurable through numerical conservation diagnostics.

Contribution Surface

- Velocity Verlet integration as the substrate's default time-stepping method.
- N-body force modeling with gravitational and Coulomb terms.
- Relativistic correction as a speed-bound mechanism.
- Elastic collision resolution for simulated entities.
- Conservation-law telemetry as runtime quality control.

Bounded Claims

Claim	Class	Proof posture	Public wording
Velocity Verlet is appropriate for long-running Hamiltonian-style simulation because its energy error is bounded under standard assumptions.	C10 / C9	Supported by known numerical analysis; needs citation pass.	"Velocity Verlet is a standard symplectic method often used when bounded long-term energy behavior is desirable."
The packet's physics engine implements N-body gravitational and electromagnetic interactions.	Potential C1	Not located in inspected repo surface.	"The canonical design targets N-body gravitational and electromagnetic interactions; the inspected repo currently verifies a tick-based 3D entity simulation layer."
Relativistic corrections prevent simulated particles from exceeding c.	C3 / potential C1	Needs implementation and test evidence.	"The design includes a relativistic speed-bound mechanism intended to constrain propagation speed."
Conservation diagnostics validate simulation quality.	C3 / C4	Methodologically plausible; thresholds need logs.	"Conservation diagnostics are proposed as quality indicators for simulation runs."

Proposed Outline

1. Introduction: physical substrate for agent simulation.
2. Numerical mechanics requirements.
3. Velocity Verlet integration and error behavior.
4. Pairwise interaction model.
5. Relativistic speed-bound policy.
6. Collision model and conservation assumptions.
7. Conservation-law telemetry.
8. Validation plan and expected diagnostics.
9. Limits and relation to later series papers.

Required Evidence Before Preprint Release

- Code references for integrator, force computation, collision resolution, and telemetry.
- Current repo-backed references: `spatium-computationis/simulation/engine.py`, `spatium-computationis/simulation/world.py`, and `spatium-computationis/simulation/routes.py` for tick-based 3D simulation.
- Energy, momentum, and angular-momentum drift plots across representative runs.
- Parameter table for time step, softening length, masses, charges, and units.
- Comparison against Euler or RK methods if claiming superior long-term behavior.
- Benchmark confirming scaling and practical particle limits.

Minimized Overlap Rule

This paper may mention agents only as physical entities requiring a numerical substrate. It should not describe the 21-agent architecture, mini-verse topology catalog, or full differential-geometry machinery except as future layers.